

Nonlinear Control Final Assignment

Haniyeh Zahra Budaqi
haniyehz.budaqi55@sharif.edu

March 3, 2026

Problem One

Consider a rigid spacecraft with a general inertia matrix $[I]$. The attitude is controlled using a set of continuously variable thrusters which produce a torque $\mathbf{u}$ about the spacecraft center of mass. The attitude is measured using a set of modified Rodrigues parameters $\sigma_{B/N}$.

a) Using **Lyapunov's direct method**, develop a globally stabilizing feedback control law to bring both the inertial attitude $\sigma_{B/N}$ and angular rates $\omega_{B/N}$ relative to an inertial orientation $\mathcal{N}$ to zero (Regulator problem). Show all work, including how to differentiate the Lyapunov function term involving $\sigma_{B/N}$.

b) Analytically prove that this control is globally asymptotically stabilizing.

Answer

a) We know the following facts:

$$\dot{\sigma} = \frac{1}{4}[(1 - \sigma^2)I + 2[\tilde{\sigma}] + 2\sigma\sigma^T]\omega$$

$$[I]\dot{\omega} = -[\tilde{\omega}][I]\omega + \mathbf{u} + \mathbf{L}$$

We choose our Lyapunov function as such to involve both $\omega_{B/N}$ and $\sigma_{B/N}$. This function should be positive definite and have continuous derivatives:

$$V(\omega, \sigma) = \frac{1}{2}\omega^T[I]\omega + 2K \ln(1 + \sigma^T\sigma)$$

Also, before differentiating the equation above, we must make this fact clear that the vectorial quantities in the equation are seen by the body frame $\mathcal{B}$ and the derivatives will also be taken with respect to the $\mathcal{B}$ frame. That being said, the Lyapunov function V is a scalar and does not care with respect to which frame it's being differentiated. Also, since $\omega = \omega_{B/N}$:

$${}^{\mathcal{B}}\frac{d}{dt}\omega = {}^{\mathcal{N}}\frac{d}{dt}\omega = \dot{\omega}$$

Now, we will take the derivative of V :

$$\begin{aligned}\dot{V} &= \frac{1}{2}\dot{\omega}^T[I]\omega + \frac{1}{2}\omega^T[I]\dot{\omega} + \frac{2K(\dot{\sigma}^T\sigma + \sigma^T\dot{\sigma})}{1 + \sigma^T\sigma} \\ &= \omega^T[I]\dot{\omega} + \frac{2K(2\sigma^T\dot{\sigma})}{1 + \sigma^2} \\ &= \omega^T[I]\dot{\omega} + \frac{4K}{1 + \sigma^2}\sigma^T\dot{\sigma}\end{aligned}$$

Now, to complete the calculation of the second part of the right hand side (the one involving $\sigma_{B/N}$, we substitute $\dot{\omega}$ with the equation we first introduced:

$$\begin{aligned}\frac{4K}{1 + \sigma^2}\sigma^T\dot{\sigma} &= \frac{4K}{1 + \sigma^2}\sigma^T\left[\frac{1}{4}[(1 - \sigma^2)I + 2[\tilde{\sigma}] + 2\sigma\sigma^T]\omega\right] \\ &= \frac{K}{1 + \sigma^2}[\sigma^T[(1 - \sigma^2)I + 2[\tilde{\sigma}] + 2\sigma\sigma^T]]\omega \\ &= \frac{K}{1 + \sigma^2}[(1 - \sigma^2)\sigma^T + 2\sigma^T[\tilde{\sigma}] + 2\sigma^T\sigma\sigma^T]\omega \\ &= \frac{K}{1 + \sigma^2}[(1 - \sigma^2)\sigma^T + 2\sigma^2\sigma^T]\omega \\ &= \frac{K}{1 + \sigma^2}[(1 + \sigma^2)\sigma^T]\omega \\ &= K\sigma^T\omega \\ &= \omega^T(K\sigma)\end{aligned}$$

Now, we rewrite $\dot{V}$:

$$\begin{aligned}\dot{V} &= \omega^T[I]\dot{\omega} + \omega^T(K\sigma) \\ &= \omega^T([I]\dot{\omega} + K\sigma)\end{aligned}$$

To guarantee Lyapunov stability, Lyapunov function derivative should be at least negative semi-definite and should be simplified to the format below:

$$\dot{V} = -\omega^T[P]\omega$$

in which $[P]$ is a positive definite matrix. That being said, the right hand side of the derived $\dot{V}$ should now equal the negative semi-definite format. Thus, we have:

$$\omega^T([I]\dot{\omega} + K\sigma) = -\omega^T[P]\omega$$

Substituting $[I]\dot{\omega}$ we will have:

$$\omega^T(-[\tilde{\omega}][I]\omega + \mathbf{u} + \mathbf{L} + K\sigma) = -\omega^T[P]\omega$$

The term $\omega^T[\tilde{\omega}][I]\omega$ cancels out and the problem assumes zero external torque ($\mathbf{L} = 0$), therefore:

$$\mathbf{u} + K\sigma = -[P]\omega$$

$$\mathbf{u} = -K\sigma - [P]\omega$$

Since the chosen Lyapunov function has all the characteristics needed, this control is proven to be globally stabilizing.

b) To prove global asymptotic stability, we will use Theorem 8.5 in the textbook [1], investigating higher derivatives of the Lyapunov function V with the set on which $\dot{V} = 0$. We already know that if $\omega = 0$ and $\sigma = 0$, $\dot{V} = 0$.

$$\dot{V} = -\omega^T [P] \omega$$

$$\begin{aligned}\ddot{V} &= -\dot{\omega}^T [P] \omega - \omega^T [P] \dot{\omega} \\ &= -2\omega^T [P] \dot{\omega}\end{aligned}$$

$$\ddot{V}(\omega = 0, \sigma = 0) = 0$$

Now, we investigate $\ddot{V}$:

$$\begin{aligned}\ddot{V} &= -2(\dot{\omega}^T [P] \dot{\omega} + \omega^T [P] \ddot{\omega}) \\ \ddot{V}(\omega = 0, \sigma = 0) &= -2\dot{\omega}^T [P] \dot{\omega} < 0\end{aligned}$$

Since the first non-zero derivative is the third derivative ($k = 3$ and odd) and also negative definite, the control is proven to be globally asymptotically stabilizing.

Problem Two

Consider the spacecraft attitude control problem when a severe spin is to be arrested. The goal of your control is to drive the body angular rates $\omega_{B/N} \rightarrow 0$. The final attitude is irrelevant, we are just happy to stop tumbling end over end. Assume no external torques are present.

c) Assuming a continuous control (without saturation), develop a globally asymptotically stabilizing control. Analytically prove these properties.

d) Next, assume that the control torque magnitudes are limited to be either 0 or $\pm u_{max}$. Develop a pure *Lyapunov optimal control* which will stop the spin. Analytically show that this control is globally asymptotically stabilizing.

e) How can this Lyapunov optimal control be modified to avoid chatter issues around zero rate condition? Analytically discuss the stability of this modified control implementation.

Answer

This is a rate regulation problem, in which the final attitude is irrelevant and the spacecraft spin rate is to be arrested.

a) Since there's no external torque present, we know that:

$$[I]\dot{\omega} = -[\tilde{\omega}][I]\omega + \mathbf{u}$$

The Lyapunov function will be the same as the one used in the problem one except for the fact that there's no need for the term involving $\sigma_{B/N}$:

$$V = \frac{1}{2}\omega^T [I] \omega$$

And for the first derivative:

$$\begin{aligned}\dot{V} &= \omega^T [I] \dot{\omega} \\ &= \omega^T (-[\tilde{\omega}][I]\omega + \mathbf{u}) \\ &= \omega^T (\mathbf{u}) = -\omega^T [P] \omega\end{aligned}$$

$$\mathbf{u} = -[P]\omega$$

Just like in problem one, we now prove that this control is globally asymptotically stabilizing.

$$\dot{V} = -\omega^T [P] \omega$$

$$\begin{aligned}\ddot{V} &= -\dot{\omega}^T [P] \omega - \omega^T [P] \dot{\omega} \\ &= -2\omega^T [P] \dot{\omega}\end{aligned}$$

$$\ddot{V}(\omega = 0) = 0$$

Now, we investigate $\ddot{V}$:

$$\begin{aligned}\ddot{V} &= -2(\dot{\omega}^T [P] \dot{\omega} + \omega^T [P] \ddot{\omega}) \\ \ddot{V}(\omega = 0) &= -2\dot{\omega}^T [P] \dot{\omega} < 0\end{aligned}$$

Since the first non-zero derivative is the third derivative ($k = 3$ and odd) and also negative definite, the control is proven to be globally asymptotically stabilizing.

No saturation was taken into account for part (a).

b) Including the saturation, we now know that:

$$\mathbf{u}_{us} = -[P]\omega$$

$$\mathbf{u}_i = \begin{cases} 0 & \|\mathbf{u}_{us_i}\| < \mathbf{u}_{max_i} \\ -\mathbf{u}_{max_i} \cdot \text{sgn}(\omega_i) & \|\mathbf{u}_{us_i}\| \geq \mathbf{u}_{max_i} \end{cases} \quad (1)$$

This leads to the Lyapunov rate function:

$$\dot{V} = -\sum \mathbf{u}_{max_i} \omega_i \cdot \text{sgn}(\omega_i)$$

Since this $\dot{V}$ is negative definite, the control law is globally asymptotically stabilizing.

c) To avoid the chatter around zero rate condition, we add the following modification:

$$\mathbf{u}_i = \begin{cases} -P_{ii}\omega_i & \|\mathbf{u}_{us_i}\| < \mathbf{u}_{max_i} \\ -\mathbf{u}_{max_i} \cdot \text{sgn}(\omega_i) & \|\mathbf{u}_{us_i}\| \geq \mathbf{u}_{max_i} \end{cases} \quad (2)$$

The corresponding Lyapunov rate function would be:

$$\dot{V} = -\sum P_{ii}\omega_i^2 - \sum \mathbf{u}_{max_i} \omega_i \cdot \text{sgn}(\omega_i)$$

Since this $\dot{V}$ is negative definite, the control law is globally asymptotically stabilizing.

References

- [1] Hanspeter Schaub and John Junkins. *Analytical Mechanics of Space Systems*. American Institute of Aeronautics and Astronautics, Inc., 2018. ISBN: 9781624105210. DOI: <https://doi.org/10.2514/4.105210>.